

# Cryo-EM structure of SRP68/72 reveals an extended dimerization domain with RNA-binding activity

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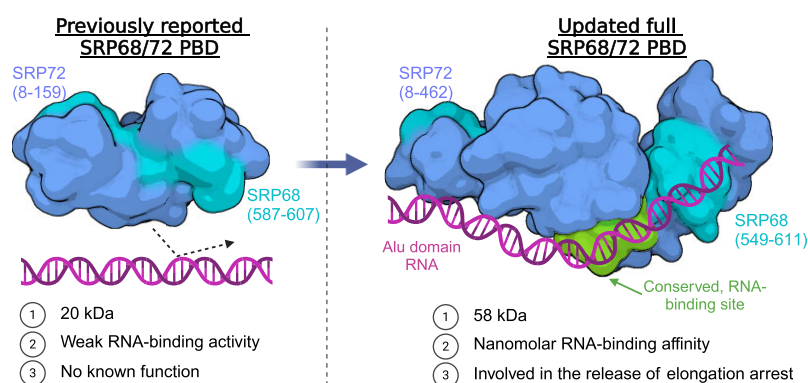
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## Abstract

The signal recognition particle (SRP) is a critical component in protein sorting pathways in all domains of life. Human SRP contains six proteins bound to the 7S RNA and their structures and functions have been mostly elucidated. The SRP68/72 dimer is the largest SRP component and is essential for SRP function. Although the structures of the SRP68/72 RNA binding and dimerization domains have been previously reported, the structure and function of large portions of the SRP68/72 dimer remain unknown. Here, we analyse full-length SRP68/72 using cryo-EM and report that SRP68/72 depend on each other for stability and form an extended dimerization domain. This newly observed dimerization domain is both a protein- and RNA-binding domain. Comparative analysis with current structural models suggests that this dimerization domain undergoes dramatic translocation upon SRP docking onto SRP receptor and eventually comes close to the Alu domain. We propose that the SRP68/72 dimerization domain functions by binding and detaching the Alu domain and SRP9/14 from the ribosomal surface, thus releasing elongation arrest upon docking onto the ER membrane.

## Graphical abstract



## Introduction

In eukaryotic cells, most membrane and secretory proteins are translocated into the endoplasmic reticulum (ER) co-translationally for modification, processing and secretion. This translocation is triggered when a cleavable N-terminal signal peptide (SP) emerges from the ribosomal exit tunnel and is recognised by the signal recognition particle (SRP), a ribonucleoprotein complex (1,2). The SRP pathway is an ancient and universally conserved mechanism in all life forms. Compared to other SRPs, the eukaryotic SRP complex is the largest with the most components, containing six proteins (SRP9, SRP14, SRP19, SRP54, SRP68 and SRP72) and a 300-nt 7S RNA. The binding of SRP to ribosome-nascent chain complexes (RNCs) arrests translation, followed by docking of the entire RNC-SRP onto the SRP receptor (SR) upon GTP binding. RNC handover and GTP hydrolysis leads to SRP and SR dissociation and insertion of the nascent chain into the Sec translocon (3), allowing resumption of translation with the newly synthesized protein being co-translationally translocated into the ER (4).

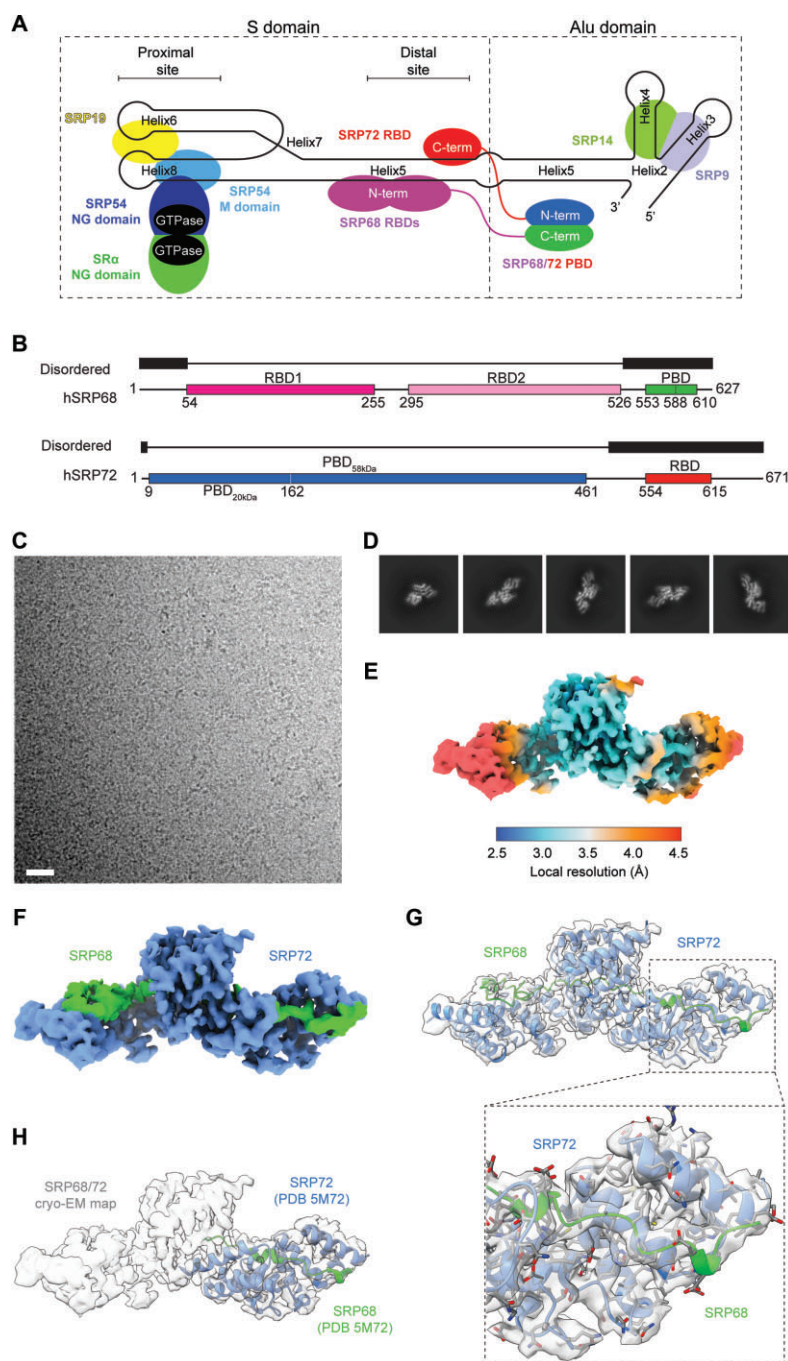
The SRP-dependent protein targeting pathway was first discovered in the 1980s using mammalian cell extracts (5–7). Since then, the domain architectures, structures, and interactions of eukaryotic SRP proteins have been heavily studied, revealing details

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**Figure 1.** Cryo-EM structure determination of human SRP68/72. **(A)** Schematic representation of the human SRP showing the six proteins and their binding sites on 7S RNA, which is separated into the S domain and Alu domain. The SRP54 NG domain forms a dimer with SR $\alpha$ . The helical segments of 7S RNA are numbered from 2–8 (36). **(B)** Human SRP68 (Uniprot: Q9UHB9) and SRP72 (Uniprot: O76094) domains that have experimentally determined structures. SRP68 and 72 PBDs and RBDs (SRP68 and SRP72 RBDs: PDB 7OBQ (17), SRP68/72 PBD: PDB 5WRV (37)) that have been observed in structural studies so far are labelled by coloured blocks. The SRP68/72 PBD reported in this study is also included. Disordered regions predicted by AF disorder (33) are shown as black rectangles above the sequences. **(C)** Representative micrograph acquired using Krios/Selectris X/Falcon 4i at  $\sim 1.7$   $\mu$ m defocus. Approx. 400–450 particles were picked from each micrograph. Scale bar: 200 Å. **(D)** Two-dimensional class averages obtained from cryoSPARC processing, revealing the presence of a rod-shaped particle in the micrographs, subsequently identified as the SRP68/72 PBD<sub>58kDa</sub>. **(E)** Local resolution estimates for cryo-EM maps of human SRP68/72 indicate flexibility of peripheral parts of the complex, leading to lower local resolution. **(F)** Cryo-EM map of the observed SRP68/72 complex coloured by protein subunit (SRP68 lime, SRP72 blue). **(G)** Cryo-EM map shown in (F) fitted with refined atomic model (top) accompanied by a close-up view of the highlighted area (bottom). **(H)** X-ray crystal structure of human SRP68/72 PBD (PDB 5M72 (35)) unambiguously fitted into our cryo-EM map.

of their functional mechanisms and giving rise to a model of SRP-dependent protein targeting. The nascent SP is recognized by the M-domain of SRP54 (8), which, together with SRP19, binds at the so-called proximal site (Figure 1A) and maintains the folding of the 7S RNA S domain (9,10). The NG domain of SRP54 forms a heterodimer with the NG domain of the SR alpha subunit (SR $\alpha$ ) upon GTP binding (11) and translocates from the proximal site to the distal site of 7S RNA S domain where GTP

hydrolysis is triggered (12). The SRP9/14 heterodimer binds the 7S RNA Alu domain and upon RNC–SRP complex formation, this sub-assembly stalls translation by occupying the elongation factor (EF) binding site on the ribosome (13,14) (Figure 1A). Nevertheless, remaining unanswered questions have so far precluded the completion of the model of SRP-dependent protein targeting. Specifically, it is unknown how GTP hydrolysis at the SRP S domain triggers the dissociation of the Alu domain from the ribosome and release of elongation arrest, and the functions of the regions of the SRP68/72 heterodimer for which no high-resolution structures exist have remained elusive.

SRP68 and SRP72 interact through the N-terminal protein binding domain (PBD) of SRP72 and a short C-terminal region of SRP68, whereas the C-terminus of SRP72 and N-terminus of SRP68 contain their RNA-binding domains (RBDs) that interact with 7S RNA at the distal site of the S domain (15,16) (Figure 1A). The SRP72 RBD is part of the complex that triggers GTP hydrolysis whilst the SRP68 RBDs are involved in either promoting or loosening elongation arrest, through an unknown mechanism (14,17). Interestingly, elastase treatment of SRP72 generated a 55 kDa species that is larger than any of the known domains (18), as the defined PBD and RBD are 18 and 8 kDa, respectively (15). Furthermore, the structure of the full length (FL) SRP68/72 dimer has never been solved by X-ray crystallography nor has it been observed in any of the cryo-EM structures of ribosome-bound SRP complexes, despite their essential role in SRP function.

In this study, we performed cryo-EM analysis using the FL SRP68/72 heterodimer, revealing an extended dimerization domain between the two proteins with a molecular weight of 58 kDa (51 kDa of SRP72 + 7 kDa of SRP68). The observed structure of the SRP72 PBD is only ~20 residues shorter than the 56 kDa domain predicted by an AlphaFold2 (AF) model, matching the elastase digested species. Further structural and functional analysis of this new PBD, which we refer to as PBD<sub>58kDa</sub>, showed that it has a highly conserved, positively charged patch with RNA-binding activity. Thus, we incorporated this new domain into the SRP targeting models derived from previous cryo-EM structures and propose that this newly identified domain controls the release of elongation arrest.

## Materials and methods

### Plasmid construction

Genes encoding human SRP68 (52–627) and SRP72 (1–671), codon optimised for *Escherichia coli* expression, were cloned into the same pDESKt plasmid, between the NotI and BamHI sites. SRP72 was expressed with a cleavable N-terminal His<sub>6</sub>-tag, followed by a 240-bp spacer and then SRP68. Both genes were transcribed from the same T7 promoter, and the spacer contains an M13 forward site for sequencing and a second ribosome binding site for SRP68 translation. The same setup was used for the SRP68/72 PBDs. SRP72 (8–172), (8–256), (8–462) wt or (8–462) mutants replace FL SRP72, and SRP68 (549–611) with a C-terminal StrepII tag (GSGWHPQFEK) replaces SRP68 (52–627) in the pDESKt plasmids. SRP72 (8–172) only and (8–256) only constructs were cloned into pLIC plasmids, and a cleavage His<sub>6</sub>-Maltose binding protein (MBP) tag was added to the N-termini instead.

The coding sequences of each RNA used in this study were cloned into pUC19 plasmid and transcribed from a T7 promoter. A HDV ribozyme was added to the end, followed by a unique BamHI site. The DNA sequences used for sAlu and s4.5S were 5'-GAGACTGAGGCTGGAGGATCGCTTGAGGAATCAAGCGAGACCCCGTCTCTAG-3' and 5'-GAGATGTTGGTTCTCCCGCAcGGAAGTGCCGGGATGTAGCTGGCA-3', respectively.

### Protein expression and purification

All proteins were expressed and purified from *E. coli* BL21 (DE3) Rosetta cells. Transformed cells were incubated in LB media at 37°C and grown until OD<sub>600</sub>~0.6, then induced with 0.5 mM isopropyl β-D-1-thiogalactopyranoside (IPTG), followed by overnight expression at 20°C.

For the SRP68/72 PBD complexes, the cells were lysed by sonication in 50 mM HEPES, pH 7.5, 0.2 M NaCl, 10 mM MgCl<sub>2</sub>, 10 mM KCl, 10% (v/v) glycerol, 1× Complete inhibitor (Roche®, COEDTAF-RO), 1 mM phenylmethylsulfonyl fluoride (PMSF), 1 mM tris (2-carboxyethyl) phosphine (TCEP). After centrifugation at 15000 × g for 30 min, the clarified lysate was incubated with Ni-nitrilotriacetic acid (Ni-NTA) beads for 1 h at 4°C. The beads were washed with lysis buffer containing 10 mM imidazole and then eluted by increasing imidazole to 500 mM. The N-terminal tag was removed by overnight dialysis with Tobacco Etch Virus (TEV) protease (produced in-house) against 20 mM HEPES, pH 7.5, 0.2 M NaCl, 10 mM MgCl<sub>2</sub>, 10 mM KCl, 10% (v/v) glycerol, 1 mM TCEP at 4°C. The uncleaved protein, His-tagged TEV protease and cleaved tags were removed by one round of reverse Ni-NTA purification. The purity of the flowthrough and washes were checked on SDS-PAGE gels before being combined and concentrated by Amicon® Ultra centrifugal filter units for RNase-free size exclusion chromatography (SEC). SEC was performed using a Cytiva® Superdex™ 75 column in 50 mM HEPES, pH 7.5, 100 mM NaCl, 10 mM MgCl<sub>2</sub>, 10 mM KCl, 10% (v/v) glycerol, 1 mM TCEP. After SDS-PAGE analysis on pre-cast gels (nvitrogen™ NuPAGE™ 4–12%, Bis-Tris, 1.0–1.5 mm, Mini Protein Gels), the clean fractions from SEC were combined and concentrated for further experiments.

For SRP72 (8–172) only and (8–256) only, the cells were lysed in 50 mM Tris, pH 8.0, 200 mM NaCl, 10% (v/v) glycerol, 0.1% (v/v) Triton X-100, 5 mM Imidazole, 1 mM TCEP, 1 mM PMSF and 1× Complete inhibitor, and dialysed against 50 mM Tris, pH 8.0, 100 mM NaCl, 1 mM TCEP overnight. The proteins were later exchanged into the same final buffer as SRP68/72 PBD complexes by SEC.

FL SRP68/72 required 1 M NaCl in the lysis buffer and was dialysed in buffer containing 400 mM NaCl. After TEV cleavage, the protein complex was then purified by cation exchange using a Cytiva® HiTrap SP HP column and elution gradient from 400 to 1000 mM NaCl. Fractions containing clean FL SRP68/72 were concentrated for RNase Free SEC, which was done with



a Cytiva® Superdex™ 200 column in 50 mM HEPES, pH 7.5, 350 mM NaCl, 10% (v/v) glycerol, 10 mM MgCl<sub>2</sub>, 10 mM KCl, 1 mM TCEP.

### Cryo-EM grid preparation

Cryo-EM samples were prepared on UltrAuFoil 1.2/1.3 300 mesh gold grids (Quantifoil Micro Tools GmbH). The grids were plasma cleaned for 50 s using a Tergeo EM Plasma Cleaner (PIE Scientific) before sample addition. 4 µl of sample containing 3 µM of SRP68/72 dimer in cryo-EM buffer (50 mM HEPES-KOH, pH 7.5, 400 mM NaCl, 10 mM MgCl<sub>2</sub>, 10 mM KCl, 0.7% (v/v) glycerol, 0.5 mM TCEP) were applied to the grid and blotted for 1–4 s at 5°C and 100% humidity using a Vitrobot Mark IV (Thermo Fisher Scientific) and then plunge-frozen in liquid ethane cooled by liquid nitrogen.

### Data acquisition

Initially, eight grids were clipped into autogrid cassettes and screened on a 200 kV Glacios cryo-transmission electron microscope equipped with a Falcon 4i direct electron detector (Thermo Fisher Scientific). Approximately 200–300 movies were collected from each grid and processed on-the-fly using cryoSPARC live (19) to assess data quality and particle orientations by 2D classification. For high-resolution structure determination of SRP68/72, selected grids were loaded into a Titan Krios G4 cryo-transmission electron microscope (cryo-TEM) (Thermo Fisher Scientific) equipped with a Falcon 4i detector, a Selectris X energy filter (operated using a slit width of 10 eV), and a 300 kV cold-FEG. Grid squares were manually selected and automatically brought to eucentric height in EPU (Thermo Fisher Scientific). Holes were automatically detected and selected under specific relative ice thickness filters suitable for SRP68/72. For data collection, the microscope was set to 245 600-fold magnification and electron micrograph movies were collected in EER format with a pixel size of 0.57 Å/pixel, a total electron exposure of 70 e<sup>−</sup>/Å<sup>2</sup>, and a defocus range of −0.5 µm to −2.0 µm. Aberration-free image shift (AFIS) and fringe-free imaging (FFI) were employed to accelerate data acquisition to approximately 550–600 movies/hour. In total, 18 601 EER format movies (966 raw EER frames/movie) were acquired.

### Image processing

SRP68/72 data were pre-processed in cryoSPARC live and cryoSPARC V3.3.1 (19), and then transferred to RELION 4.0 (20) for further processing and 3D reconstruction (Supplementary Figure S1). Final refinement was performed in RELION 5.0 beta using BLUSH regularisation (21). In cryoSPARC live (Supplementary Figure S1), raw movies were fractioned into 50 frames and motion corrected with 2× binning. The contrast transfer function (CTF) of each motion-corrected micrograph was fitted within cryoSPARC live. Total beam-induced specimen motion, in-frame motion, relative ice thickness, and the quality of CTF fitting were inspected to remove poor-quality micrographs, which led to 6 269 micrographs being rejected and 12 332 micrographs being accepted. Particles were blob-picked using the circular and elliptical blob options (80–140 Å diameter), and then extracted using a box size of 180 × 180 pixels. The quality of the dataset was assessed by streaming 2D classification and 3D homogeneous refinement of particles from four good 2D classes. Using a previous reconstruction of SRP68/72 obtained from refinement of a smaller Glacios dataset against a rod-like initial volume as an initial reference, a final count of 574 507 particles were refined to 3.3 Å resolution, as assessed by Fourier Shell Correlation (FSC) at the 0.143 cut-off (22).

To maximize the retrieval of high-quality particles from the dataset, three particle picking strategies were employed in cryoSPARC V3.3.1 (19) (Supplementary Figure S1). Blob picking particles were obtained by re-classification of the blob-picked particles from the cryoSPARC live session into 200 2D classes using a batch size of 200 particles (i.e. 40 000 particles being considered in each iteration). Eventually, 447 244 blob-picked particles from good 2D classes were retained (particle set (i); particle sets are identified in Supplementary Figure S1). In addition to blob-picking, particles were also extracted using template picking and the deep-learning based Topaz picking (23) with an extraction box size of 180 × 180 pixels. For template picking, two picking jobs were set up. The first job utilized 2D averages generated from 2D classification of blob picking particles as templates for particle picking; the second job utilized slightly different 2D averages generated by the first job as templates for a second round of picking. 416 925 and 385 755 template-picked particles were selected after 2D classification (particle sets (ii) and (iii)). Topaz picking was trained on a set of ~120 000 high-quality particles arising from the best 2D classes generated by blob picking. Topaz picking and subsequent 2D classification yielded 478 210 good particles (particle set (iv)). Selected particle sets (i)–(iv) extracted by the three different picking algorithms in cryoSPARC V3.3.1, plus selected particles extracted during the cryoSPARC live session (particle set (v)), incorporating one additional 2D class, bringing the number of particles retained from the live session to 699 932, were joined and duplicate particles were removed. Subsequently, the combined particles were re-extracted and re-centred, and duplicates generated during these processes were removed again. Finally, 1 198 158 particles from all picking strategies were exported and converted to a RELION-compatible \*.star file using a conversion program in the PYEM package (24) with adjustment to abide by RELION conventions, and were then imported into RELION 4.0 (Supplementary Figure S1).

In RELION 4.0 (20) (Supplementary Figure S1), 18 580 EER format movies successfully processed in cryoSPARC live were imported. During motion correction with a binning factor of 2 by the RELION implementation of a MOTIONCOR2-like algorithm (25), every 14 raw EER frames were grouped into one fraction, resulting in 69 movie fractions and 1.01 e<sup>−</sup> Å<sup>−2</sup> per fraction. Particles were re-extracted from the motion-corrected micrographs using a 180 × 180 pixel extraction box using coordinate information contained in the imported particles from cryoSPARC V3.3.1. In total, 1 198 158 particles were extracted and subjected to masked 3D auto-refinement using a previous SRP68/72 reconstruction as an initial model. Particles were then subjected to a 3D classification with no alignment (regularization parameter  $\tau = 20$ ). At this stage, only a small number of non-particle images (stripe artifacts, ice edges) were removed. The remaining 1 196 301 particles from five good 3D classes were selected and auto-refined to 3.5 Å resolution. To further improve the map quality, those particles underwent Bayesian polishing

(26) and were then auto-refined to 3.3 Å resolution. Subsequently, those particles were subjected to an additional round of alignment-free 3D classification (regularization parameter  $\tau = 20$ ). 242 040 particles from the best 3D class were selected, refined, and subjected to Bayesian polishing. Those polished particles were then 3D auto-refined and post-processed to 2.9 Å.

Because the map quality was not yet suitable for reliable atomic model building and refinement, likely due to residual heterogeneity (Supplementary Movie 1), this particle set was further processed using BLUSH regularization in RELION 5 beta (21). After re-refinement, an alignment-free 3D classification (regularization parameter  $\tau = 32$ ) was performed, upon which a single high-quality class comprising 38 514 particles was selected and again refined to 3.0 Å resolution (Supplementary Figure S2). BLUSH regularization was used for both refinement and classification and led to clear improvement in the appearance of the map, rendering it suitable for reliable model building and refinement.

### Model building and refinement

To build the SRP68/72 PBD<sub>58kDa</sub> model, an AF prediction (27) was fitted into the post-processed 3.0 Å-resolution cryo-EM density maps using UCSF CHIMERA (28), and then transferred to COOT (29) for model building. The flexible parts of the predicted model that were not visible in the density maps were manually deleted (Supplementary Figure S3), followed by several rounds of manual protein model rebuilding. The models were then refined by real-space refinement in PHENIX (30). To avoid overfitting in the less-well resolved peripheral parts of the map (see below), the highest spatial frequency considered during refinement was set to 3.2 Å resolution. The resulting models were validated using MOLPROBITY (31) as implemented in PHENIX. Refinement statistics are provided in Supplementary Table S1.

### Creation of figures

Molecular graphics figures were rendered in UCSF CHIMERA (28), UCSF CHIMERA X (32) and PyMOL (The PyMol molecular graphics system, Schrödinger LLC).

### RNA probe preparation and labelling

pUC19 plasmids containing the gene for each RNA were linearized by overnight BamHI digestion. On the next day, the DNA was purified by phenol-chloroform extraction followed by chloroform extraction and ethanol precipitation. The DNA was then resolubilized in MilliQ water (MQW) and used as template (0.5 mg per 10 ml reaction) for *in vitro* transcription by mixing with 40 mM Tris pH 8.0, 2 mM Spermidine, 10 mM NaCl, 25 mM MgCl<sub>2</sub>, 0.01% (v/v) Triton X-100, 7.5 mM of each NTP, 40 mM DL-dithiothreitol (DTT), 0.01 mg/ml Pyrophosphatase, 0.02 U RiboSafe RNase inhibitor (Bioline, Cat. No.: BIO-65028) and 0.1 mg/ml T7 RNA polymerase (produced in-house). The reaction was incubated at 37°C for 3 h. For ribozyme self-cleavage, the transcription products were replenished with additional 20 mM MgCl<sub>2</sub>, then incubated at 95°C for 2 min and immediately on ice for 3 min. The heating cycles were repeated three times in total. After mixing with 2× RNA loading dye (80% (v/v) Formamide, 1× TBE, Xylene cyanole FF and Bromophenol Blue), the sample was heated at 95°C for 2 min before loading onto a 6% (v/v) denaturing urea gel consisting of 7 M urea, 6% (19:1) pre-mixed acrylamide-bisacrylamide and 1× TBE. The gel was pre-run until reaching ~50°C, and the RNA sample was resolved on gel by running at 25 W for 2–3 h at room temperature. The RNA band was visualised by UV shadowing, the correct band excised, and eluted by crushing the gel and incubating in MQW with shaking overnight at 4°C. The resulting suspension was filtered to remove the gel pieces and exchanged into MWQ and concentrated using ethanol precipitation. The RNA was stored at –80°C until use.

RNAs used for EMSA were labelled using 5' EndTag™ DNA/RNA Labeling Kit (Vector Laboratory, MB-9001) and Cy5-maleimide (Kerafast Inc., MA, USA) by following the manufacturer's protocol. Briefly, 0.6 nmol of RNA was mixed with alkaline phosphatase in universal reaction buffer (supplemented in the kit) and incubated for 30 min at 37°C. T4 polynucleotide kinase and ATPγS (supplemented in the kit) were then added to the mixture, followed by 30 min incubation at 37°C. The mixture was then reacted with Cy5-maleimide dissolved in DMSO for 2 h at room temperature in the dark. The RNA was purified by phenol-chloroform extraction and concentrated by ethanol precipitation. The labelled RNA was redissolved in water and stored at –80°C until use.

### Electrophoretic mobility shift assay (EMSA) and microscale thermophoresis (MST)

Each reaction contained 33 nM of Cy5-labelled RNA, 50 mM HEPES, pH 7.5, 100 mM NaCl, 3 mM MgCl<sub>2</sub>, 10 mM KCl, 5% (v/v) glycerol, 0.01% (v/v) IGEPAL® CA-630 and 10 µg/ml (~400–500 nM) yeast tRNA (Sigma-Aldrich, R8508). SRP68/72 PBDs were prepared by serial dilution using SEC buffer and incubated with the mixture for 30 min on ice, before loaded onto 5% (for RNA < 100 nt) or 6% (for RNA ≥ 100 nt) (19:1) 1× TBE gels. The gel was run in 0.5 × TBE buffer at 1 W/10 ml gel for 1.5 h at 4°C and then imaged on an FLA-9000 laser scanner. The fraction of bound RNA was measured by densitometry using ImageJ and plotted using nonlinear regression, specific binding with Hill slope in GraphPad Prism version 10.0.2 (GraphPad Software, Boston, Massachusetts USA, <http://www.graphpad.com>) with  $B_{\max}$  constricted between 0–1. The fluorescent RNA stuck at the bottom of the gels were not included in measurement or calculations. All EMSA were repeated for at least three times and the most representative gels are shown in the figures. The number of replicates is indicated in the figure legends.

The MST reactions were prepared in the same buffer except the SRP68/72 PBD<sub>58 kDa</sub> was prepared by 2:3 serial dilutions. The samples were loaded into standard capillary tubes before undergoing MST in a Monolith™ NT.115 instrument (NanoTemper Technologies GmbH, München, Germany). Thermophoresis was conducted at 100% LED power, and 40% MST power. Thermophoresis data from three technical replicates were then analysed with MO Affinity Analysis software v2.3 and the dis-

sociation constants ( $K_D$ ) were determined using the  $K_D$  binding model based on the Langmuir binding isotherm and a 1:1 interaction model:

$$f(C) = F_{\text{norm}(\text{target})} + \frac{(F_{\text{norm}(\text{complex})} - F_{\text{norm}(\text{target})}) \times \left( C + C_{\text{target}} + K_D - \sqrt{(C + C_{\text{target}} + K_D)^2 - 4 \times C \times C_{\text{target}}} \right)}{2 \times C_{\text{target}}}$$

where  $f(C)$  is the fraction bound at a given ligand concentration 'C'.  $F_{\text{norm}}$  is the normalized fluorescence signal which is defined by the ratio of relative fluorescence of the selected hot region and relative fluorescence of cold region (1.0 s before MST starts).  $F_{\text{norm}(\text{target})}$  is the normalized fluorescence signal of unbound RNA and  $F_{\text{norm}(\text{complex})}$  is the normalized fluorescence signal of bounded RNA to the PBD<sub>58kDa</sub>.  $C_{\text{target}}$  is the final concentration of the RNA in the assay.

### Differential scanning fluorimetry (DSF)

DSF profiles of SRP72 or SRP68/72 PBDs were obtained using an Qiagen™ Rotor-Gene Q PCR instrument. SYPRO Orange dye was mixed with buffer to reach 50-fold dilution of the original stock solution and mixed with proteins to give a final protein concentration of 0.3 or 0.6 mg/ml and a total sample volume of 20 µl. Protein deficient control wells were similarly adjusted with buffer. Reactions were performed in triplicate. Changes in fluorescence were measured for temperatures ranging from 25 to 90°C at a ramp rate of 1°C/min. The relative fluorescence units (RFU)/first derivatives of RFU versus temperature were calculated and plotted using GraphPad Prism version 10.0.2 (GraphPad Software, Boston, MA, USA, <http://www.graphpad.com>).

## Results and discussion

### Cryo-EM structure determination

SRP68 and SRP72 are the largest protein components of the eukaryotic SRP, yet only part of their structures have been experimentally determined (68% of SRP68 and 22% of SRP72). Interestingly, both proteins are predicted to be largely ordered (33) (Figure 1B). Therefore, we sought to characterize the structure and function of the FL SRP68/72 complex. SRP68 (52–627) and SRP72 (1–671) were co-expressed and purified with approximately 1:1 stoichiometry according to SDS-PAGE analysis (Supplementary Figure S4A), and concentrated fractions were used for extensive crystallographic trials. We found that the FL SRP68/72 complex was not amenable to crystallization and required the use of cryo-EM imaging to solve its structure.

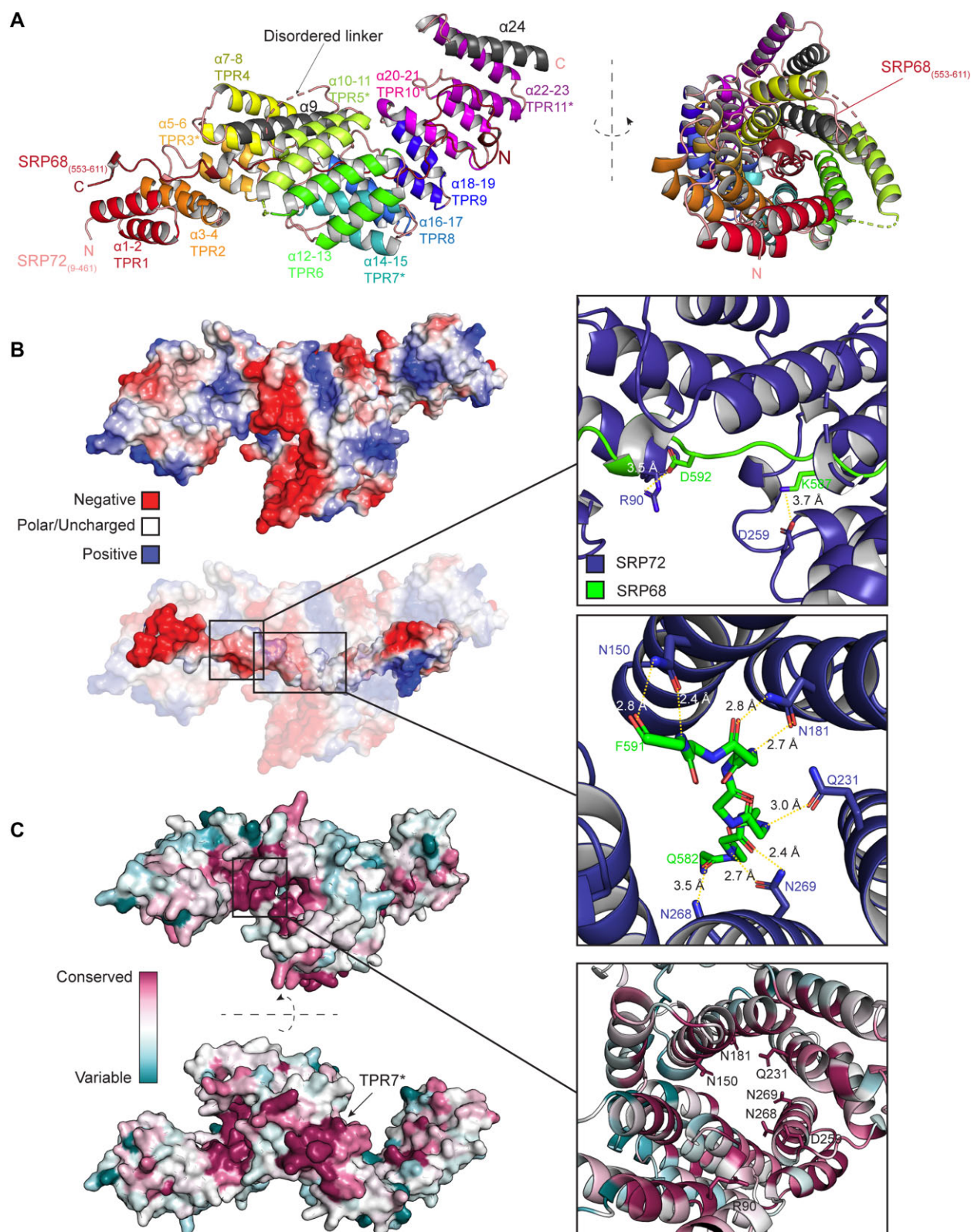
The analysis of initial SRP68/72 cryo-EM data by 2D classification revealed the presence of a rod-like particle that was too small to account for the entire SRP68/72 complex (Figure 1C, D). Due to the size and flexibility of this rod-like module, a high-resolution structure suitable for model building remained elusive using data collected on a Glacios cryo-TEM, even from very large datasets comprising ~20 000 movies. This preliminary analysis revealed that the particle resolved in our cryo-EM maps was not only very small for cryo-EM standards (~55–60 kDa of mass resolved), but also flexible (Figure 1E, Supplementary Movie 1), complicating structure determination. To address these challenges, we used a Krios G4 cryo-TEM equipped with a cold-FEG, Selectris X energy filter, and Falcon 4i direct electron detector to collect the best possible data and employed various image processing approaches, including strict 3D classification in RELION (20) and 3D flexible refinement in cryoSPARC (34) to obtain cryo-EM maps that could be interpreted reliably in conjunction with AF modelling (27). We found that the cryo-EM maps from BLUSH regularization implemented in RELION 5 (21) provided the best balance of resolution, map quality, and map anisotropy and used a 3.0 Å-resolution map generated by this approach (Figure 1F, G, Supplementary Figures S1 and S2) for atomic model building.

Fitting of existing structures into our SRP68/72 cryo-EM map revealed that part of the density could be explained by the X-ray crystal structure of the SRP68/72 PBD (PDB 5M72 (35) and PDB 5WRV (36)), while the remainder could not be accounted for by any of the known structures (Figure 1H). Combining our cryo-EM map with AF predictions (Supplementary Figure S3) reaffirmed the identification of our structural model as a larger SRP68/72 PBD with a substantially extended form compared to previous interpretations (35). The molecular weight of this extended PBD, composed of SRP72 residues 9–461 and SRP68 residues 553–608, is roughly 58 kDa. We therefore refer to this module as PBD<sub>58kDa</sub> to distinguish it from the previously characterised 20 kDa SRP68/72 PBD (PDB 5M72 with SRP72 (10–159) and SRP68 (587–607) (35), which we refer to as PBD<sub>20 kDa</sub>.

### Cryo-EM structure of SRP68/72 shows a larger protein binding domain

The cryo-EM results revealed an extended PBD of SRP68/72 composed of more interactions between SRP68 and SRP72 than in the previous structures (35,37). Compared with the first structure of the SRP68/72 PBD<sub>20 kDa</sub> (PDB 5M72 (35), C<sub>α</sub> RMSD of 0.9 Å), PBD<sub>58 kDa</sub> covers residues 9–461 of SRP72, which includes 15 additional α-helices beyond the nine α-helices present in PBD<sub>20 kDa</sub>, nearly tripling the size of the known SRP68/72 PBD (Figure 1B). These 24 helices coil around the SRP68 PBD in a right-handed super-helical manner to form 11 tetratricopeptide repeats (TPRs) (Figure 2A). A disordered linker (SRP72 S79–T109) was found between a pair of parallel α-helices, α9 and α10 (Figure 2A), thus breaking the continuous pattern of anti-parallel helix-turn-helix motifs in the structure. Residues 462–510 of SRP72 were not observed in the structure and are predicted to form three extra helices by AF (Supplementary Figure S4B), suggesting that the very C-terminal α24 observed in our cryo-EM reconstruction might be the first half of another TPR. The quality and resolution of the cryo-EM density deteriorate towards this region, indicating structural flexibility (Figure 1E, Supplementary Movie S1). Among the 11 observed TPR motifs, 5 of them (TPR3\*, TPR5\*, TPR7\*, TPR10\* and TPR11\*, labelled with asterisks) are considered non-classical TPRs since they lack the signature TPR residues or contain mismatching residues between the observed structure and the predicted TPR sequence





**Figure 2.** Structural analysis of the PBD<sub>58 kDa</sub> of the SRP68/72 heterodimer. **(A)** Structure of SRP68/72 PBD<sub>58 kDa</sub>. SRP72 (9–461) and SRP68 (553–608) are shown in cartoon. Helices of each TPR or TPR\* pair are coloured the same, whereas unpaired helices are in black. **(B)** Electrostatic surface potential map of SRP72 (*top left*) and SRP68 (*bottom left*). Electrostatic and hydrogen bonding interactions between the two proteins are shown in the boxes on the right. **(C)** Conservation map of SRP72 PBD<sub>58 kDa</sub>. Annotation of conserved residues was generated by ConSurf server (<https://consurf.tau.ac.il/>). All residues directly involved in H-bonding in (B) are found to be >95% conserved (dark purple residues, *right*), whereas residues around TPR7\* also form a second cluster of conserved residues (*bottom left*).

(Supplementary Figure S5). These non-classical TPRs are not uncommon in TPR-containing proteins and might contribute to the substrate-binding groove by presenting a surface different to the TPR consensus sequence.

The cryo-EM structure also shows a longer SRP68 PBD (553–608), which is mainly an extended peptide except for a short helix between L593 and N596. The SRP68 PBD extends from SRP72  $\alpha$ 24 towards  $\alpha$ 1 with their interface stabilized by at least three types of interactions. Near the N-terminus of SRP72, SRP68 F590 and F600 interact with the hydrophobic pockets formed by SRP72  $\alpha$ 2–3, and  $\alpha$ 8–9, respectively (35). There are four salt bridge pairs formed between SRP68 K606:SRP72 D44, SRP68 D592:SRP72 R90, SRP68 H597:SRP72 E113 and SRP68 K587:SRP72 D259 (Figure 2B, top right). A network of H-bonding interactions is observed between the main chain of SRP68 P583–A594 and the side chains of SRP72 residues on the interior surface of TPR5\*–TPR8 (Figure 2B, bottom right). Interestingly, mutations to residues involved in SRP68/72 interactions, such as D44E, D259G, N268T and Q381R in SRP72 as well as P583H/L, P585L and F590L in SRP68, have been associated with cancer (Supplementary Table S2 and Supplementary Figure S6A–S6D). Additional cancer-associated mutations have been reported for the PBD<sub>20 kDa</sub> (37), suggesting a possible correlation between the stability and function of the SRP68/72 PBD.

The analysis of the distribution of conserved residues on SRP72 using the ConSurf server (consurf.tau.ac.il) highlights two main clusters of conserved residues on SRP72, with conservation higher than 95% across 323 CD-HIT unique homologs (Figure 2C and Supplementary Figure S5B). The first cluster is on the inner surface formed by TPR5\*–TPR8 (Figure 2C, top left), with SRP72 residues involved in electrostatic interactions and the H-bonding network with SRP68 being highly conserved (dark purple residues, Figure 2C, right). This suggests that the interactions between the SRP68/72 PBDs might be conserved for most eukaryotic SRP complexes. The second cluster of conserved residues is exposed on the exterior surface of SRP72 TPR7\*, ranging from I252 to N290 (Figure 2C, bottom left). This cluster contains several positively charged residues exposed on the surface of the SRP72, especially between I271–N290 (Supplementary Figure S5B), which can form a functional RNA binding surface that could not be otherwise achieved by classical TPRs.

In summary, the structural analysis of the FL SRP68/72 heterodimer revealed the PBD<sub>58 kDa</sub>, which is much larger than previously identified. The SRP68 PBD is only an extended peptide but fits into the concave groove of the SRP72 solenoid through an intricate interaction network. The SRP72 in PBD<sub>58 kDa</sub>, on the other hand, is a superhelix made of TPR-like domains. Among them, TPR7\* is the most conserved TPR with a possible RNA binding function.

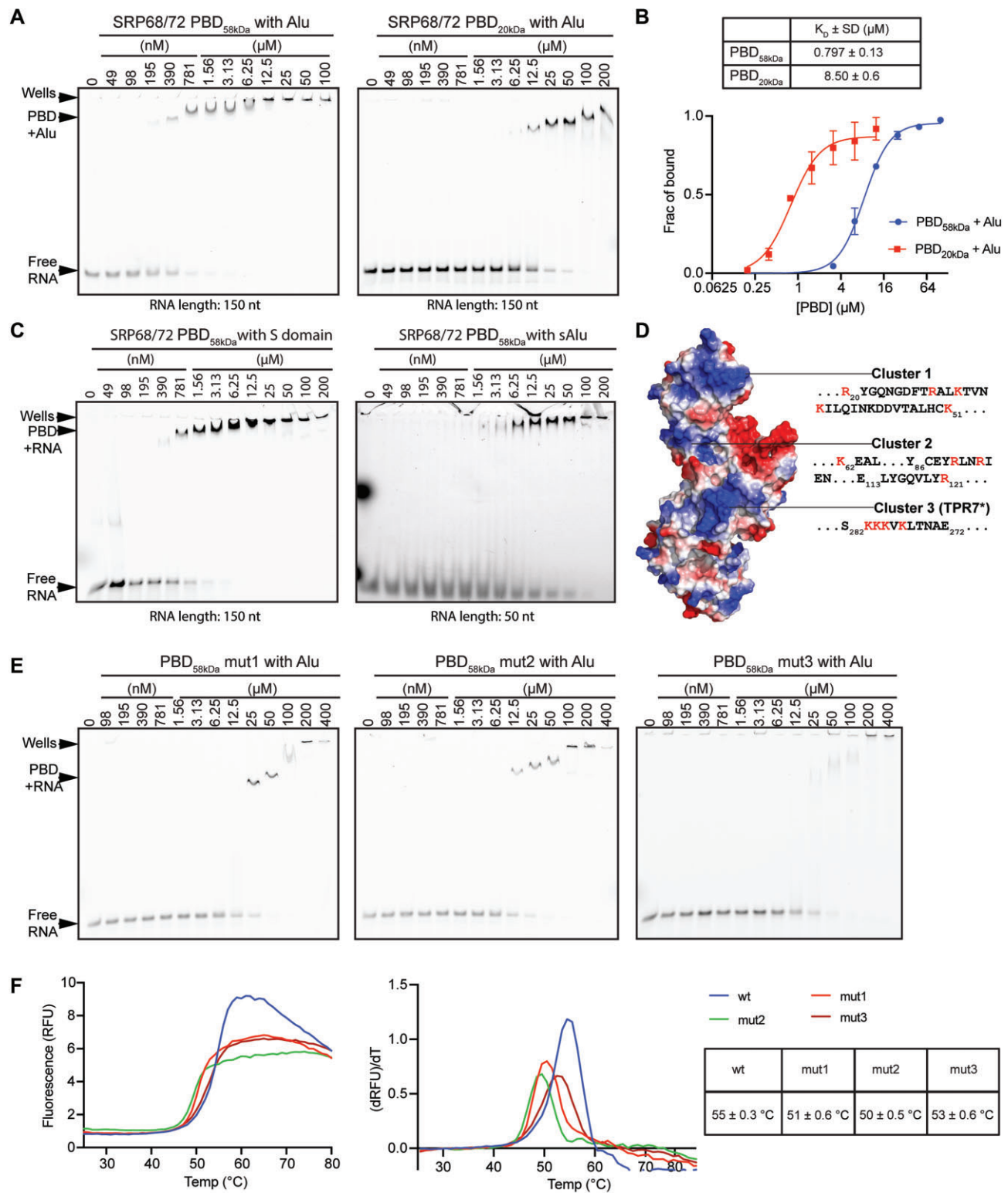
### SRP68/72 PBD<sub>58 kDa</sub> is a non-sequence specific RNA-binding domain.

To test the hypothesis that the SRP68/72 PBD<sub>58 kDa</sub> has RNA-binding activity, purified PBD<sub>58 kDa</sub> complex was used in EMSAs with Cy5-labelled Alu domain RNA. RNA band shifts could be observed starting at 195 nM SRP68/72 concentration with an estimated  $K_D$  of  $\sim$ 800 nM (Figure 3A, left), indicating that the SRP68/72 PBD<sub>58 kDa</sub> is also an RNA-binding domain. In contrast, the SRP68/72 PBD<sub>20 kDa</sub> complex binds the Alu domain RNA with a much weaker affinity, with band shifts starting at 3.25  $\mu$ M and an estimated  $K_D$  of  $8.5 \pm 0.6 \mu$ M (Figure 3A, right and Figure 3B). The affinity between the SRP68/72 PBD<sub>58 kDa</sub> and the Alu domain is estimated to be 10-fold tighter than that of SRP68/72 PBD<sub>20 kDa</sub>, possibly because the latter lacks the conserved TPR7\*.

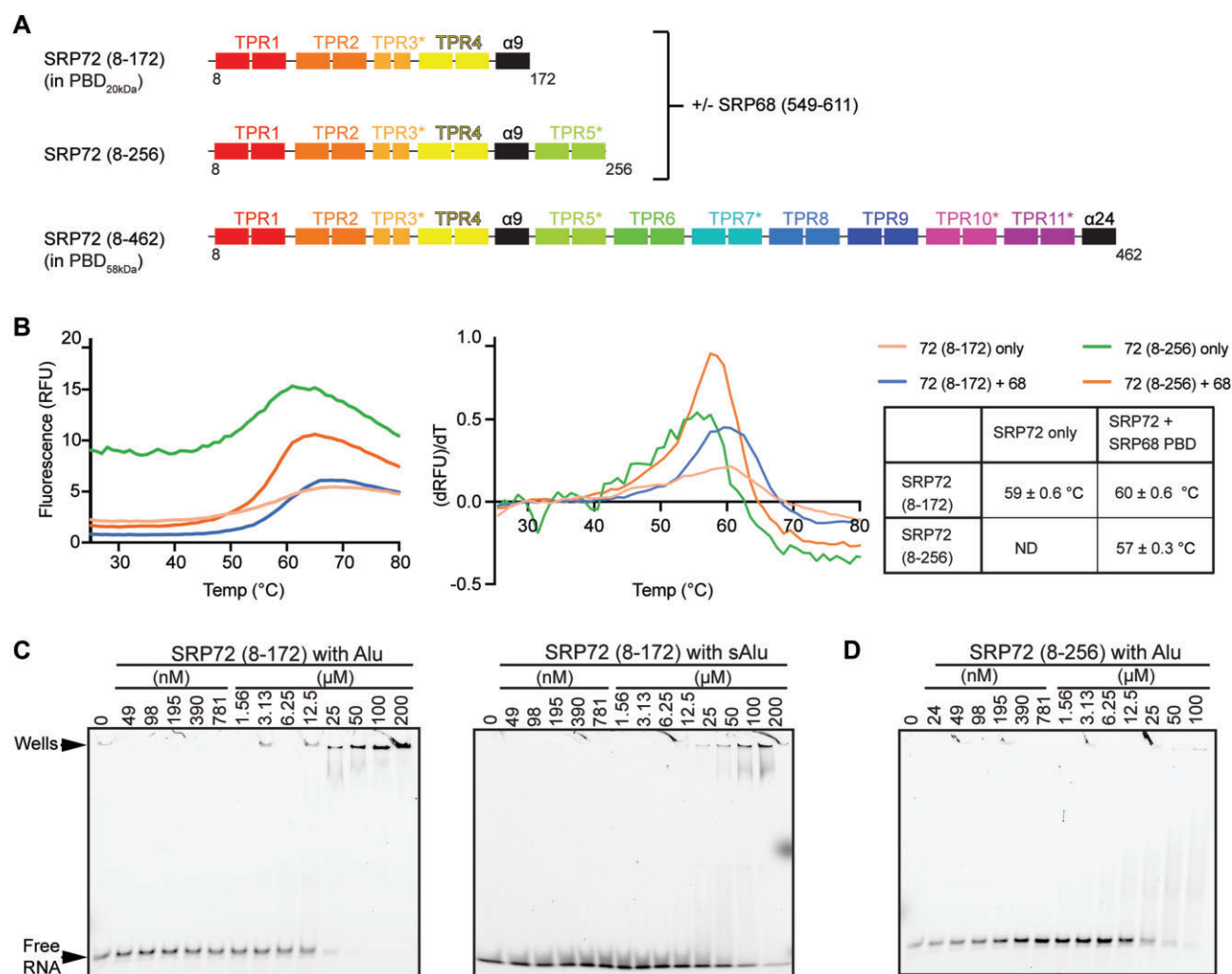
To determine if the SRP68/72 PBD<sub>58 kDa</sub> binds RNA in a sequence-specific manner like the well-characterized SRP68 and SRP72 RBDs, we performed EMSAs with the 7S RNA S domain, FL 7S RNA (Figure 3C, left), and the prokaryotic 4.5S RNA (Supplementary Figure S7A). The initial RNA shifts of all three RNAs start at 390 nM with an estimated  $K_D$  between 0.6–1  $\mu$ M (Supplementary Figure S7B). To confirm the observed results from EMSA, microscale thermophoresis (MST) was used to quantitatively determine the  $K_D$  of PBD<sub>58 kDa</sub> with Alu and S domain RNA (Supplementary Figure S7C), as  $3.2 \pm 0.7$  or  $3.9 \pm 0.8 \mu$ M, respectively. The affinities for Alu and S domain are relatively consistent with the estimation from EMSA (both in low micromolar range), indicating the lack of sequence specificity for the RNA-binding activity of PBD<sub>58 kDa</sub>. Furthermore, PBD<sub>58 kDa</sub> displays no structural preference for the RNA substrates, as the 4.5S RNA, Alu/S domain, or 7S RNA fold into various combinations of long dsRNA regions with small bulges and/or branched loop regions (Supplementary Figure S7D). Since the SRP68/72 PBD<sub>58 kDa</sub> displayed no striking specificity among the RNAs ranging from 100–300 nts, shorter RNA substrates were tested. Using a smaller version of the Alu domain (sAlu with 50 nts) and the 4.5S RNA (s4.5S with 40 nts), the SRP68/72 PBD<sub>58 kDa</sub> binding weakens and both RNAs start shifting at 3  $\mu$ M (with an estimated  $K_D$  of  $\sim$ 9  $\mu$ M) (Figure 3C, right and Supplementary Figure S7A–S7B). Compared to the longer RNAs (Alu domain: 150 nt, S domain: 150 nt, 7S FL: 300 nt, and 4.5S: 110 nt), sAlu and s4.5S binding is 10-fold weaker, indicating that SRP68/72 PBD<sub>58 kDa</sub> has an RNA length preference.

Upon analysis of the electrostatic potential map of the SRP68/72 PBD<sub>58 kDa</sub>, two other clusters of positive residues can be observed on the same surface side as TPR7\*, which might explain the RNA length preference of the SRP68/72 PBD<sub>58 kDa</sub>. We will refer to these sets of residues as clusters 1, 2 and 3 (Figure 3D). Cluster 1 is composed of positive residues between SRP72 R20 to K50 with five of them being  $>60\%$  conserved (Supplementary Figures S5B and S7C). Residues of cluster 2 are divided between TPR3\* and TPR4, whereas the charges of cluster 3 are mainly attributed to the positively charged residues on TPR7\*. To test the effect of each cluster in RNA-binding, several basic residues in each cluster were mutated into polar residues without disrupting the protein solubility and stability (Figure 3D and Supplementary Figure S8A). The resulting mutants (termed mut 1, 2 and 3) were expressed and purified as an SRP68/72 complex and their RNA-binding activity was measured via EMSAs. As expected, all three mutants have reduced affinities for the Alu domain, with RNA band shifts starting at 25  $\mu$ M, when compared with the SRP68/72 PBD<sub>58 kDa</sub> wt. Among them, mut 3 shows the weakest binding without displaying distinct bands as observed with wt, mut 1 and 2 (Figure 3E). Thus, all three clusters contribute to RNA-binding, with cluster 3 (in the TPR7\*) being the most important for the binding affinity. Clusters 1 and 2, on the other hand, are likely to be responsible for the interactions between SRP68/72 and the stability of the superhelical structure, as indicated by their lower melting temperatures when compared with





**Figure 3.** SRP68/72 PBD<sub>58kDa</sub> binds to dsRNA. **(A)** EMSA gels of increasing concentrations (0–100 μM) of SRP68/72 PBD<sub>58kDa</sub> (left) and (0–200 μM) of PBD<sub>20kDa</sub> (right) with Cy5-labelled Alu domain RNA. The free RNA and the shifted RNA bands are indicated by arrows. **(B)** Binding curves depicting % bound RNA with increasing PBD<sub>58kDa</sub>/PBD<sub>20kDa</sub> concentrations. Three independent EMSAs were performed for each protein, and error bars represent standard deviation (SD).  $K_D \pm SD$  are listed in the table above. **(C)** EMSA gels of increasing concentrations of SRP68/72 PBD<sub>58kDa</sub> (0–200 μM) with Cy5-labelled S domain or sAlu RNA. **(D)** Electrostatic surface map of PBD<sub>58kDa</sub> showing three clusters of positive residues on the same side as TPR7\*. Basic residues in TPR7\* form cluster 3. Residues to be mutated into polar residues are coloured in red. **(E)** EMSA gels of increasing concentrations of SRP68/72 PBD<sub>58kDa</sub> mut 1–3 (0–400 μM) with Cy5-labelled Alu domain RNA. **(F)** DSF melting curves of SRP72 PBD<sub>58kDa</sub> wt and mutants. Each trace is an average of three technical replicates. Data are represented in normalized relative fluorescence units (RFU) (left), and the first derivative of RFU with respect to temperature (right). Melting temperatures (mean ± SD) corresponding to the highest first derivatives of each protein are listed in a table.



**Figure 4.** Comparing the stabilities of SRP72 PBD constructs with or without SRP68 co-expression. **(A)** Schematic diagrams showing SRP72 constructs comprising residues 8–172, 8–256, or 8–461. TPRs and TPR\*s are coloured in the same scheme as in Figure 2B. **(B)** DSF melting curves of SRP72 PBD<sub>20kDa</sub> with or without SRP68 PBD<sub>58kDa</sub> co-expression. Each trace is an average of three technical replicates. Data are represented in normalized relative fluorescence units (RFU) (*left*), and the first derivative of RFU with respect to temperature (*right*). Melting temperatures (mean ± SD) corresponding to the highest first derivatives of each protein are listed in a table. **(C)** EMSA gels of increasing concentrations (0–200 μM) of SRP72 (8–172) with Cy5-labelled Alu RNA (*left*) or sAlu RNA (*right*). **(D)** EMSA gel of increasing concentrations (0–100 μM) of SRP72 (8–256) with Cy5-labelled Alu RNA.

mut 3 and SRP68/72 PBD<sub>58 kDa</sub> wt (Figure 3F). Interestingly, only K51N (38) and R93K (39) in cluster 1 and 2 were found associated with cancer in the published variants on Catalogue of Somatic Mutations in Cancer (COSMIC) website.

Therefore, SRP68/72 PBD<sub>58 kDa</sub> is not only larger than initially defined but also harbours a new function as an RNA-binding domain that interacts with dsRNA. The SRP68/72 PBD<sub>58 kDa</sub> displayed neither structure- nor sequence-specificity for RNA interaction partners but showed nanomolar-low micromolar range affinities for dsRNAs longer than ~100 nts. At least three clusters of positively charged residues on the PBD<sub>58kDa</sub> surface are involved in RNA-binding, with cluster 3 around TPR7\* being the main contributor, consistent with the hypothesis that TPR-like domains could be substrate binding site on a protein formed by TPR arrays.

### The SRP68 PBD stabilizes the extended solenoid of SRP72 PBD

The SRP68/72 PBD<sub>58kDa</sub> observed in this study redefines and triples the known size of the SRP68/72 PBD<sub>20kDa</sub> and its constituent protein interfaces (Figures 1G, H and 2A). The presence of the disordered linker between (S79–T109) may reduce the overall rigidity of the SRP72 PBD superhelix in the absence of SRP68 (Figure 2A) and make a longer SRP72 solenoid unstable. To test this hypothesis, two shorter constructs of the SRP72 PBD comprising residues 8–172 and 8–256 were generated and expressed either with or without the SRP68 PBD (549–611). SRP72 (8–172) covers α-helices 1–9 as well as the disordered linker, whereas SRP72 (8–256) contains two additional helices (i.e. TPR5\*) (Figure 4A). All four constructs can be expressed in soluble form (Supplementary Figure S8B). However, purified SRP72 (8–256) slowly precipitates if not bound to SRP68. This low stability was confirmed with differential scanning fluorimetry (DSF) assays (Figure 4B) in which the trace for SRP72 (8–256) has a higher background fluorescence level and precludes any further experiments with this construct. In contrast, SRP72 (8–256) when co-

expressed with SRP68 PBD shows a clear melting point at 58°C and SRP72 (8–172) with or without SRP68 present exhibits melting temperatures of 59 and 60°C, respectively (Figure 4B, left). These results indicate the SRP72 structure is unstable on its own for any construct beyond the disordered linker between  $\alpha 9$  and  $\alpha 10$ . However, this adverse effect can be avoided by the presence of the SRP68 PBD, possibly through the extended sequence with a relatively high proportion of proline residues (14.3%, comparing to 6.3% on average for human proteins). Interestingly, cancer-associated mutations were also found at these proline residues (P583H/L and P585L) (Supplementary Table S2 and Supplementary Figure S6C and D) (40,41), but *in vitro* studies of these mutations are impaired due to the instability of SRP72 PBD alone. Considering the low nanomolar affinity of SRP72 (1–163) and SRP68 (509–614) from PBD<sub>20kDa</sub> (37), a similar or stronger affinity would be expected for the longer SRP68/72 PBD<sub>58kDa</sub> construct.

The absence of SRP68 also influences the RNA-binding activity of the SRP72 PBD. Compared to the SRP68/72 PBD<sub>20kDa</sub> (Figure 3A), SRP72 (8–172) shifts Alu domain RNA in a similar pattern with smeared bands, indicating potential differences in binding dynamics (Figure 4C, left). However, SRP72 (8–172) still displays substrate length preferences, as higher concentrations are required for shifting sAlu RNA (Figure 4C, right). The smeary effect was also observed with SRP72 (8–256) (Figure 4D and Supplementary Figure S8B), except the RNA bands were only partially shifted by SRP72 (8–256), possibly due to the low stability of the protein.

Overall, our data indicate that SRP68 and SRP72 depend on each other for stabilization as observed in the melting curve experiments and by EMSA. However, these observations do not clarify the significance and function of the large SRP68/72 dimerization domain. SRP68 and SRP72 have their own unique and specific RNA-binding domains that can independently bind to 7S RNA (15,16), whilst the SRP68/72 PBD<sub>58 kDa</sub> RNA binding activity was never observed before. To obtain further insight into the possible functions of the PBD<sub>58kDa</sub> by molecular modelling, we proceeded to analyse the PBD<sub>58 kDa</sub> in the context of the current model of RNC-SRP docked onto SR.

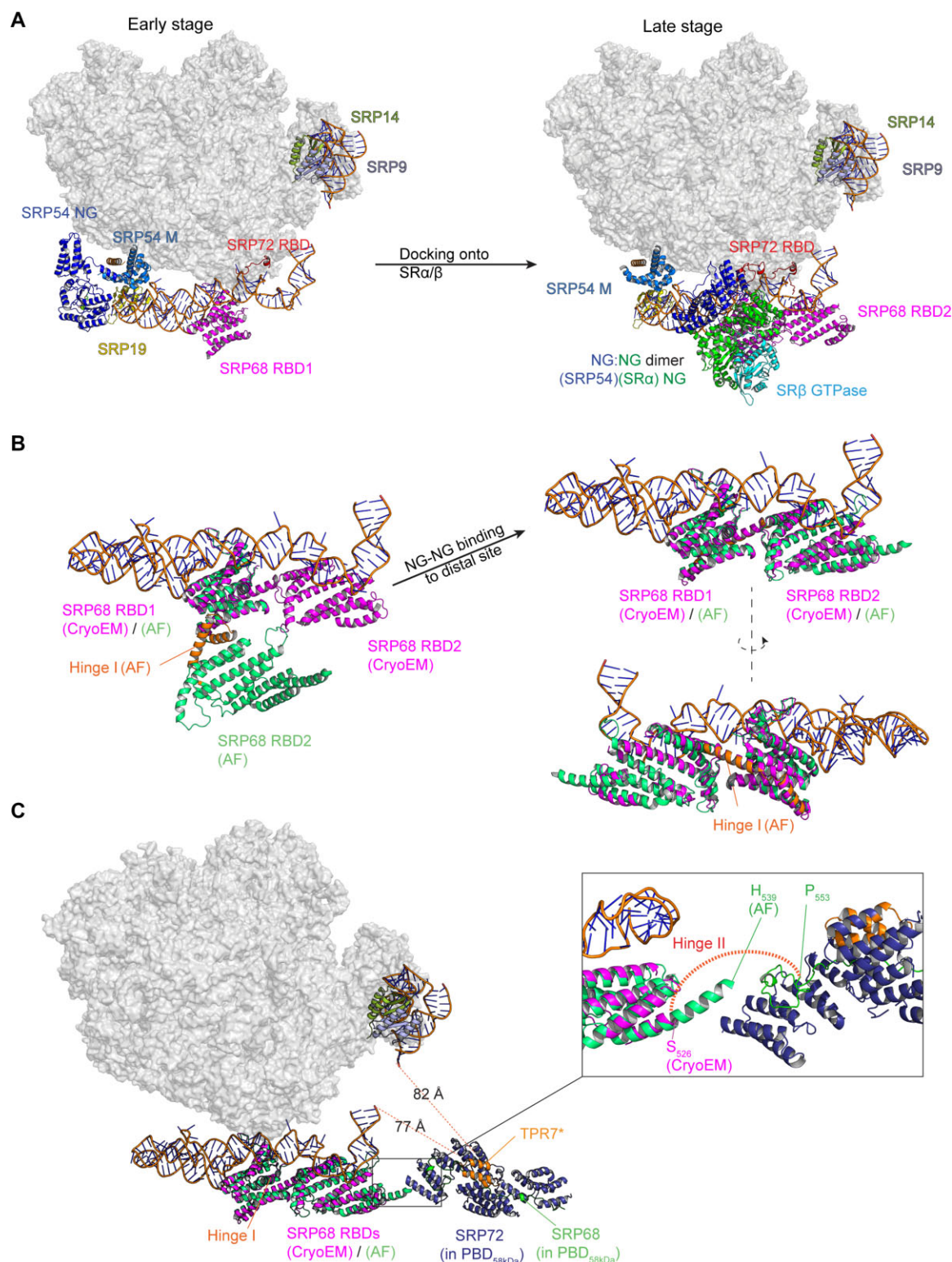
### A new model of SRP68/72 function based on comparative modelling with existing RNC-SRP-SR structures

The current model of SRP-dependent protein targeting proposes that only the first RBD (RBD1) of SRP68 binds to 7S RNA at an early stage (PDB 7NFX (42)) (Figure 5A, left). SRP68 RBD2 is thought to be flexible until the heterodimer formed by the SRP54 and SR $\alpha$  NG domains and the SR complex undergo a conformational change and re-locate to the 7S RNA distal site. Once the late-stage complex is formed and SRP68 RBD2 is placed in closer proximity to the S domain of the 7S RNA, their interaction can be observed (PDB 7OBQ (17)) (Figure 5A, right). Both RBDs of SRP68 are observed in the late RNC-SRP-SR complex and in the AF-predicted SRP68/72 structure, yet in different conformations (Figure 5B, left). The structural difference between the AF model and the experimentally determined structure is likely due to the lack of other components of the late-stage complex in the AF model. The presence of the 7S RNA and the other proteins in the late-stage complex provides the correct binding interactions for the SRP68 RBD2, accounting for the presence of a flexible linker region, S255–E295, between the SRP68 RBD1 and RBD2. Indeed, after rotating this linker region in the AF model, both RBDs and the linker overlay with the cryo-EM structure. Therefore, S255–E295 will be referred as Hinge I (Figure 5B, right). The AF model of RBD2 contains a longer C-terminus than observed in the RBD2 experimentally (PDB 7OBQ), ending with H539 (Figure 5C, green), which is only 14 residues away from P553 (N-term) of SRP68 in PBD<sub>58kDa</sub>. Sequence alignment of SRP68 from ten species reveals a relatively less conserved region between RBD2 and PBD<sub>58kDa</sub> (Supplementary Figure S8D) (L533–P553), suggesting a less sequence specific functionality for this region than its flanking domains. We thus refer to the SRP68 linker region (S526–P553) connecting RBD2 to PBD<sub>58kDa</sub> as Hinge II and considering the currently available experimental structures and steric constraints imposed by the presence of other components and linker lengths, we propose that SRP68/72 would most likely be positioned in the region shown in Figure 5C. The presence of two linkers (Hinge I and II) in SRP68 explains the difficulties in observing the PBD<sub>58kDa</sub> density in cryo-EM structures of ribosomal complexes. The structural flexibility afforded to the SRP68/72 PBD<sub>58 kDa</sub> by the presence of Hinge II allows it to sample a considerable conformational space, including possible interactions with the RNA stem from the Alu domain. Hence, SRP68/72 could play a role in sensing and coordinating the conformational changes in the S domain, near the polypeptide exit tunnel on the ribosome, with the Alu domain at the elongation-factor binding site.

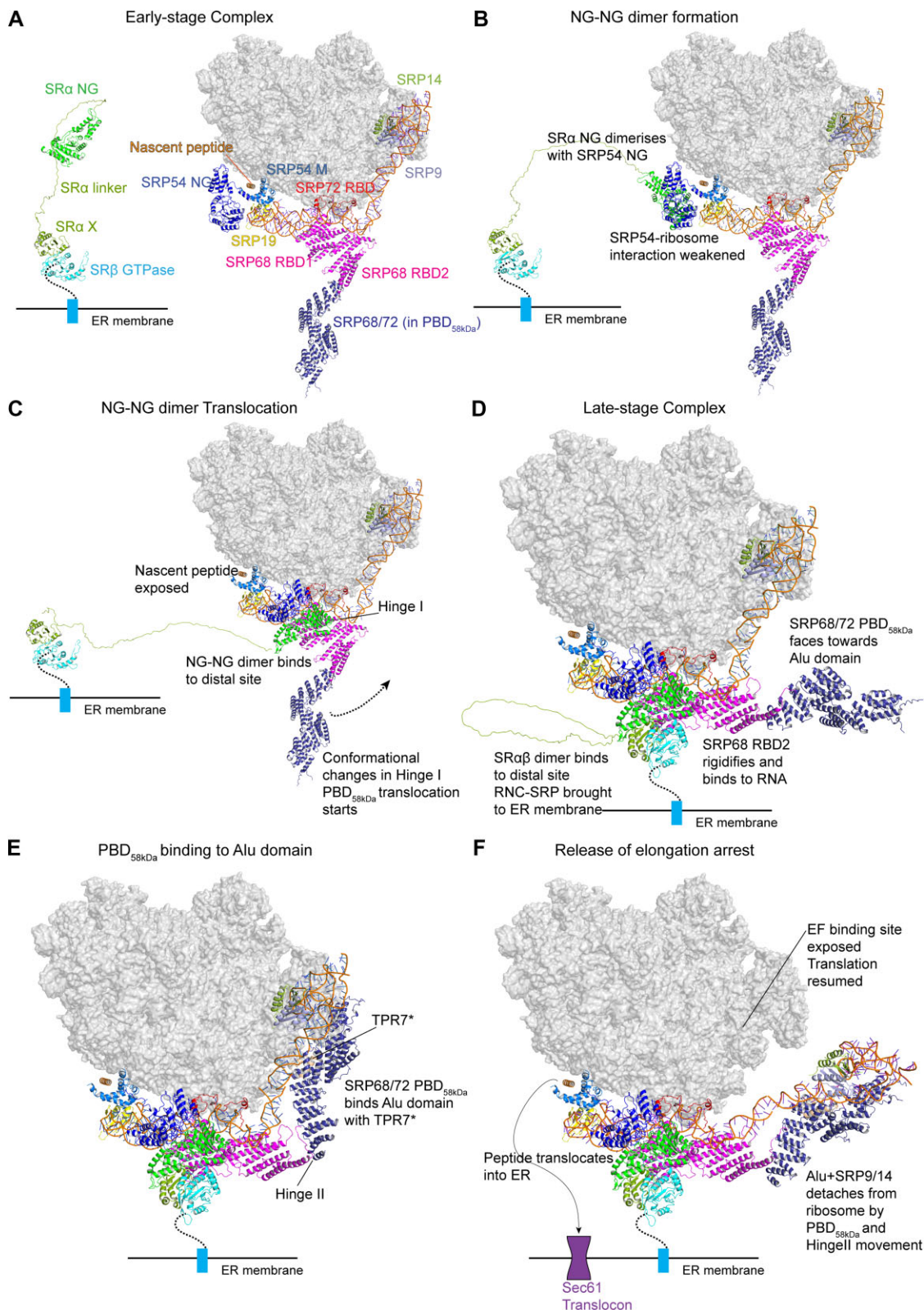
This positioning of SRP68/72 PBD<sub>58 kDa</sub> near the RNA stem leading to the Alu domain in the late-stage complex combined with its RNA-binding activity suggest that it could play a role in releasing elongation arrest. Elongation arrest is achieved by the interactions of positively charged patches of SRP9/14 with small-subunit ribosomal RNA, as well as the contacts of the Alu domain RNA with ribosomal proteins of the large subunit in the EF binding site (14,44,45). Pausing translation during elongation is a critical step for proper delivery of proteins destined to the secretory pathway since the nascent peptide has only a finite time window during which it can successfully insert into the translocon at the ER membrane. If the Alu domain and SRP9/14 were released prematurely before docking onto the translocon or if the nascent peptides were to be synthesized too far, the efficiency of translocation would be greatly reduced (46). Thus, the timing for the release of elongation arrest must be tightly controlled, and previous studies have shown that the release requires both SRP68/72 and SR $\alpha$ . SRP containing SRP68/72 treated with N-ethylmaleimide (40), a sulfhydryl specific alkylating agent, was still able to induce elongation arrest but failed to release it, even though the SRP particle can still bind SR (47). On the other hand, the SR complex is able to weaken the SRP-RNC interaction and subsequently remove the SRP-mediated translation arrest in the absence of the ER membrane (48). This early characterisation of SRP68/72 and SR, combined with our cryo-EM data, provides us with a more complete model of the docking of RNC-SRP onto the SR and the release of elongation arrest mechanism.

Our model for SRP68/72 activity in Alu domain release is illustrated in Figure 6. In the early-stage complex (Figure 6A), SRP binds to the RNC complex. All known structures are shown, including the SR $\alpha$  NG domain connecting to the X domain via a 200-residue linker, and the SR $\beta$  GTPase, which is anchored to the ER membrane. When SR $\alpha$  and SRP54 form the NG heterodimer





**Figure 5.** Proposed location of PBD<sub>58 kDa</sub> during the RNC-SRP docking process. **(A)** SRP68 RBD2 rigidifies after the SRP54-SR $\alpha$  NG heterodimer relocates from the proximal site to the distal site. In the pre-docking/early-stage complex model (*left*), SRP19/54, SRP68, and SRP72 RBDs are based on PDB 7NFX (42); RNC, 7S RNA and SRP9/14 are based on PDB 6FRK (12). In the pre-handover/late-stage structure (*right*), the SRP54 NG domain, SR $\alpha$ / $\beta$ , SRP72 RBD and SRP68 RBDs are adopted from PDB 7OBQ (17). **(B)** Conformational changes of Hinge I in SRP68 RBDs when the SRP54-SR $\alpha$  NG heterodimer binds to the distal site. SRP68 RBDs (magenta) observed in PDB 7OBQ (43) are superposed with the counterpart predicted by AF. When the RBD1s of both structures are aligned, the RBD2s are mismatched (*left*). However, rotating flexible loops in Hinge I (orange) leads to alignment of both RBDs (*right*), leading to the conformation observed in the late-stage complex (17). **(C)** Proposed location of PBD<sub>58kDa</sub> in the late-stage complex. AF-predicted SRP68 RBD2 (dark purple, V296–H539) overlays with RBDs observed in the cryo-EM structure (magenta, V296–S526, from PDB 7OBQ (17)), with the C-terminus extending towards the Alu domain and connecting with SRP68 in our cryo-EM structure of the PBD<sub>58kDa</sub>. The linker between S526 and K552 is termed Hinge II and represented by a red dash line. TPR7\* of SRP68/72 PBD<sub>58kDa</sub> in this position faces towards Alu domain RNA, ~77 and 80 Å away from the ends of the portions of the RNA resolved in the RNC-SRP cryo-EM structures, respectively.



**Figure 6.** Mechanistic model of RNC-SRP docking and release of elongation arrest. **(A)** SRP binding induces elongation arrest. Upon recognition of nascent SP through the SRP54 M and NG domains, SRP binds the RNC and Alu + SRP9/14 blocks the EF binding site, pausing translation and causing elongation arrest. SRP68/72 locate close to the distal site of S domain RNA, with RBD1 of SRP68 and RBD of SRP72 interacting with the RNA directly, whereas the rest of the dimer (RBD2 and PBD<sub>58 kDa</sub>) remains flexible. **(B)** The SRα NG domain forms a heterodimer with SRP54 NG domain, weakening the interaction between SRP54 and the ribosome. **(C)** The NG heterodimer translocates from the proximal site to the distal end, exposing the nascent SP and inducing a conformational change in Hinge I. **(D)** SRα X domain and SRβ are brought to distal site by the SRα linker, thus docking the RNC-SRP complex onto ER membrane and rigidifying Hinge I. PBD<sub>58kDa</sub> is now positioned in proximity to the Alu domain. **(E)** The flexible linker between SRP68 RBD2 and PBD<sub>58kDa</sub> allows TPR7\* to bind the dsRNA region of the Alu domain. Other positive patches on PBD<sub>58kDa</sub> enhance the interaction. **(F)** Hinge II relaxation peels Alu + SRP9/14 from ribosomal surface, freeing the EF binding site. Elongation arrest is released, and the nascent peptide will be synthesised and translocated into the ER.



complex upon GTP binding, a conformational change is induced in the SRP54 NG domain, weakening its attachment to the ribosome (Figure 6B) (17). The NG heterodimer then translocates to the distal site of the 7S RNA, exposing the M domain and the nascent peptide for cargo transfer to the translocon (Figure 6C). Meanwhile, this movement brings the RNC-SRP complex to the ER membrane through the SR $\alpha$  linker and allows the SR $\alpha$  X-SRP $\beta$  GTPase dimer to dock onto the distal site of the 7S RNA, forming the late-stage complex (Figure 6D). The binding of the SR $\alpha\beta$  dimer triggers conformational changes in SRP68, enabling RBD2 to bind the 7S RNA and assume the conformation visualised in the cryo-EM structure of the late-stage RNC-SRP complex (17). The SRP68/72 PBD<sub>58 kDa</sub> is then in close proximity to the Alu domain (Figure 6D). The flexible Hinge II between SRP68 RBD2 and PBD<sub>58 kDa</sub> enables the latter to approach and bind to the exposed Alu domain (Figure 6E), thus releasing elongation arrest by detaching or loosening Alu + SRP9/14 from the elongation binding site on ribosomal surface and resuming translation (Figure 6F). Since GTP hydrolysis happens only when the SRP54 and SR $\alpha$  heterodimer is located at the distal site of the S domain in the presence of SR $\beta$ , this model provides a mechanism for how the effects induced by SR binding, RNC handover to the translocon, and GTP hydrolysis can be sensed at the Alu domain, thereby controlling elongation arrest. An animated version of this model is provided in [Supplementary Movie S2](#).

The existence of bacterial SRP systems that contain a 6S RNA with an Alu domain but not SRP68/72 raises the question of why the eukaryotic SRP system requires SRP68/72 to achieve release of elongation arrest. We hypothesise that this requirement may stem from the higher efficiency of the eukaryotic SRP9/14 + Alu domain complex to stall translation, thus enabling tighter control of translational stalling and release compared with prokaryotic complexes containing the 6S RNA. The Alu domain in the 6S RNA from *Bacillus subtilis* SRP delays the expression of proteins containing a signal sequence more efficiently than the *E. coli* SRP, which lacks the Alu domain in its 4.5S RNA (49). The Alu domain of *B. subtilis* functions similarly to its eukaryotic counterpart by blocking the EF-binding site (49). However, the *B. subtilis* Alu domain is protein-free and binds the ribosome via loose RNA-RNA interactions instead of the tight protein-protein and protein-RNA interactions observed in the eukaryotic SRP (49,50). The evolution of a higher-affinity SRP9/14 + Alu domain in eukaryotes may explain why eukaryotic SRP requires contributions from additional proteins, including SRP68/72, to control the release of elongation arrest while these additional protein components are dispensable in bacterial systems with lower-affinity and lower-efficiency Alu domains.

In summary, by combining early characterization and functional studies with all current structural and activity data, we propose that the function of the large dimerization domain in SRP68/72 is to bind and detach the Alu + SRP9/14 module from the ribosomal surface. According to our model, the conformational change in SRP68 is triggered by SR binding, which explains how the release of elongation arrest can only be triggered after RNC-SRP complex docking onto the ER, thereby preventing failed cargo translocation caused by over-synthesized peptide.

## Data availability

The cryo-EM map and refined coordinate model have been deposited in the Electron Microscopy Data Bank (EMDB) with accession code EMD-18674 and in the Protein Data Bank (PDB) with accession code PDB: 8QVW, respectively.

## Supplementary data

[Supplementary Data](#) are available at NAR Online.

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## Conflict of interest statement

A.F.K. and A.K. are employees of Thermo Fisher Scientific, the manufacturer of the electron microscope used in this study.

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